EE1-05 Energy Conversion. Comments on students' answers (2019)

Question 1

a) Marks were typically lost for not using the general expression for the flux, e.g. assuming a uniform field or a field parallel to normal.
b) Marks were typically lost for not starting with Maxwell's equations, for using the field of a point charge to justify the answer, or for using specific paths (lines and circular segments).
c) The typical error was to omit the minus sign in front of the derivative for the potential.
d) Most students got the correct answer, but some answered incorrectly that inductance depends on the current.
e) A common error was to explain why permeability is needed at all (not asked). Also some students explained at length how ferromagnets magnetise, which was not asked.
f) A common error was to state that the values of the components are measured in the tests. However, it is the power, current, and voltage that are measured, and the component values are calculated from those measurements. (Details of how these calculations are done were not asked and received no marks). Sometimes, students drew equivalent circuits that included the secondary coil, which made their answers either incorrect or incomplete.

Question 2

a) A number of students failed to explain here why the field lines are circles but used this result in part b). Some students stated here that the field strength outside of the toroid is zero, which is correct in principle, but cannot be deduced without calculations.
b) The largest error here was to attempt to find the field using Biot-Savart's law, which always led to incorrect results (often, even with incorrect dimensions).
c) A number of students stated that the bolt would somehow be in electrical contact with the transformer windings. If it were correct, the bolt could not be used at all as it would be life-threatening. Some students correctly stated that the bolt and the framework would form a short circuit but failed to explain its significance. The framework itself already forms a short circuit, and many of the incorrect answers would have sapplied to the metallic framework even without the bolt.

Question 3

A big error, leading to marks lost in a), c) and d) was failure to notice that the left and right halves of the line were charged with opposite signs. In a) a number of students performed integration incorrectly; in c) a number of students incorrectly reasoned about the direction of the field from one of the segments which led them to state the field was directed vertically and thus answered d) incorrectly. Common mistakes in e) were incorrect integration and attempting to use Gauss's law.